Blepharocheilodontic Syndrome 1 (BCDS1): A Comprehensive Disease Characteristics Report

Disease: Blepharocheilodontic Syndrome 1 (BCDS1) **MONDO ID:** MONDO:0054740 **OMIM:** #119580 **Orphanet:** ORPHA:1997 **Causal gene: CDH1** (E-cadherin; HGNC:1748; NCBI Gene 999) **Synonyms:** BCDS1; blepharo-cheilo-dontic syndrome 1; CDH1-related blepharocheilodontic syndrome; Elschnig syndrome **Category:** Mendelian, autosomal dominant

Summary

Blepharocheilodontic syndrome 1 (BCDS1) is an ultra-rare, congenital, autosomal dominant malformation syndrome caused by **heterozygous pathogenic variants in CDH1**, the gene encoding the epithelial cell–adhesion molecule **E-cadherin**. The syndrome is defined by a clinical triad of (1) eyelid malformations (lagophthalmos, ectropion of the lower eyelids, euryblepharon, distichiasis), (2) cleft lip and/or cleft palate, and (3) dental anomalies, most notably oligodontia with conical teeth and incisal notching. BCDS1 is the CDH1-related form of the disorder; a closely related second type, BCDS2, is caused by truncating variants in CTNND1 (encoding p120-catenin), the direct binding partner of E-cadherin, and together the two genes define a single "cadherin–catenin complex" developmental disorder.

The core mechanism is dysfunction of the **E-cadherin–p120-catenin adherens-junction complex**. Unlike the loss-of-function CDH1 variants that drive hereditary diffuse gastric cancer (HDGC), the missense CDH1 variants found in BCDS1 act largely in a **dominant-negative** manner, impairing the adhesion function of the cadherin–catenin complex during the morphogenesis of the eyelids, lip/palate, and teeth. Functional studies in zebrafish, mouse (conditional Cdh1/Ctnnd1 deletion; Ctnnd1 knockout), and Xenopus recapitulate features of the syndrome and confirm the central role of E-cadherin–p120-catenin interaction in craniofacial development. Because E-cadherin sequesters β-catenin at the cell membrane, its dysfunction also links the disorder to Wnt signaling and epithelial-to-mesenchymal transition (EMT) control, though the precise downstream developmental cascade in human tissue is largely inferred.

Clinically, BCDS1 features are **structural, congenital, and non-progressive**, with normal life expectancy; management is entirely **symptomatic and multidisciplinary** — eyelid reconstructive surgery, cleft lip/palate repair, and prosthodontic/orthodontic dental rehabilitation. A clinically important and still-contested nuance is a **variant-dependent cancer risk**: at least one BCDS family developed diffuse gastric cancer from a germline CDH1 variant, while the recurrent c.760G>A (p.Asp254Asn) orofacial-cleft allele has been shown NOT to predispose to hereditary cancer. This mandates variant-specific genetic counseling, with consideration of International Gastric Cancer Linkage Consortium (IGCLC) surveillance protocols for families carrying HDGC-class alleles.

Key Findings

Finding 1 — BCDS1 is caused by heterozygous, dominant-negative CDH1 variants

BCDS1 results from heterozygous pathogenic variants in **CDH1 (E-cadherin)**. In the defining molecular cohorts, heterozygous CDH1/CTNND1 variants were identified in 15 of 17 BCDS index patients (PMID: 29348693), and an earlier study found five CDH1 heterozygous missense mutations plus three CTNND1 truncating mutations across 11 patients from 8 families (PMID: 28301459). Critically, functional experiments established that the CDH1 variants in BCDS do not behave like classic cancer-associated loss-of-function alleles:

> "Functional experiments in zebrafish and human cells showed that the CDH1 variants impair the cell adhesion function of the cadherin-catenin complex in a dominant-negative manner. Variants in CDH1 have been linked to familial hereditary diffuse gastric cancer and invasive lobular breast cancer; however, no cases of gastric or breast cancer have been reported in our BCDS cases." (PMID: 29348693)

The two distinct variant classes — **CDH1 missense** (dominant-negative) and **CTNND1 truncating** (loss-of-function/haploinsufficiency) — both converge on the same adhesion complex:

> "We identified five CDH1 heterozygous missense mutations and three CTNND1 heterozygous truncating mutations leading to loss-of-function or haploinsufficiency." (PMID: 28301459)

Additional pathogenic CDH1 missense alleles have since broadened the spectrum, e.g., c.1198G>A (p.Asp400Asn) located in the extracellular cadherin-type repeats, co-segregating in a Chinese BCD family and classified likely pathogenic under ACMG criteria (PMID: 36552944).

Finding 2 — The clinical triad: eyelid malformations, cleft lip/palate, and dental anomalies

BCDS is defined by a recognizable triad:

> "Blepharocheilodontic syndrome (BCDS) consists of lagophthalmia, ectropion of the lower eyelids, distichiasis, euryblepharon, cleft lip/palate and dental anomalies and has autosomal dominant inheritance with variable expression." (PMID: 29348693)

The **dental phenotype** is severe and quantified: in a dedicated craniofacial/dental study, all patients exhibited oligodontia with a mean of 13.7 missing permanent teeth (range 7–17), severe skeletal Class III malocclusion (vs controls, P = 0.001), reduced anterior lower face height (P = 0.027), taurodontism, conical teeth, and incisal notching (PMID: 28181393):

> "All patients exhibited oligodontia (mean number of missing permanent teeth 13.7, range 7-17)" (PMID: 28181393)

The **periocular phenotype** (lower lid ectropion, euryblepharon, lagophthalmos) is often functionally significant and surgically demanding (PMID: 39442179). BCDS overlaps clinically with isolated lagophthalmos syndrome, supporting a phenotypic continuum (PMID: 15930907). Oligodontia also places BCDS within a broader differential of oligodontia-associated syndromes alongside hypohidrotic ectodermal dysplasia, Axenfeld-Rieger, Witkop, Ellis-van Creveld, and oculofaciocardiodental syndromes (PMID: 38132417).

Finding 3 — Gastric cancer risk is variant-dependent and remains contested

The initial BCDS reports argued that BCD CDH1 variants are a distinct, non-oncogenic class, having observed no gastric or breast cancer among patients (PMID: 29348693; PMID: 28301459). This picture was complicated by subsequent reports:

> "We report the first case of diffuse gastric cancer in an individual with familial blepharocheilodontic syndrome (BCD) due to a germline CDH1 likely pathogenic variant." (PMID: 32302040)

This cancer risk was reinforced in a confirmatory follow-up (PMID: 32483297). Conversely, a focused study of the recurrent cleft allele found no cancer predisposition:

> "The Recurrent E-Cadherin (CDH1) Mutation c.760G>A Causes Orofacial Clefts but Does Not Predispose to Hereditary Cancer" (PMID: 39596675)

Interpretation: cancer risk in CDH1-BCDS is not uniform and must be assessed at the level of the specific variant, not the syndrome label.

Finding 4 — E-cadherin–p120-catenin adhesion dysfunction underlies BCDS; animal models recapitulate features

The mechanistic centerpiece is the physical interaction between E-cadherin and p120-catenin (CTNND1). Conditional genetic ablation of either partner reproduces the phenotype:

> "Conditional deletion of Cdh1 and Ctnnd1 in various animal models induces features reminiscent of BCD syndrome and underlines critical role of the E-cadherin-p120ctn interaction in eyelid, craniofacial, and tooth development." (PMID: 28301459)

CDH1 missense mutations caused accelerated E-cadherin degradation, and Ctnnd1 knockout mice recapitulate the cleft lip/palate (CL/P) spectrum, with the genes enriched in embryonic oro-palatal epithelia (PMID: 29805042):

> "recapitulation of the CL/P spectrum in Ctnnd1 knockout mice support a causative role in CL/P pathogenesis" (PMID: 29805042)

Complementary work linked p120-catenin to epithelial integrity/turnover and mesenchymal roles via conditional Ctnnd1 deletion in mice and CRISPR/Cas9 targeting in Xenopus (PMID: 32196547).

Finding 5 — The CTNND1 (BCDS2) phenotype is expanding beyond the classic triad

While BCDS1 (CDH1) typically presents the classic triad, the CTNND1-related form (BCDS2) carries a broader, multi-system phenotype:

> "presenting with craniofacial dysmorphisms including cleft palate and hypodontia, as well as congenital cardiac anomalies, limb dysmorphologies and neurodevelopmental disorders" (PMID: 32196547)

Eight novel protein-truncating CTNND1 variants (6 de novo) were found in 13 participants from 9 families, prompting the proposal of a "broader velocardiofacial-like syndrome." Newer reports add genitourinary anomalies:

> "the patient exhibited a duplex kidney, ureterocele, and a bicornuate uterus-phenotypic traits not previously associated with BCD syndrome" (PMID: 40028877)

Prenatal CTNND1 cases have described hydrocephalus and major fetal cardiac malformations (PMID: 39487033; PMID: 37274823), and a Chinese pedigree documented a novel c.83delG (p.A29Rfs*55) CTNND1 variant in non-simplex BCDS (PMID: 36584998).

Finding 6 — E-cadherin, β-catenin sequestration, Wnt signaling, and EMT

E-cadherin is a transmembrane glycoprotein forming adherens junctions; its best-characterized downstream link is sequestration of β-catenin, connecting the BCDS lesion to Wnt signaling and EMT:

> "In normal cells, E-cadherin exerts its tumour suppressing role mainly by sequestering β-catenin from its binding to LEF (Lymphoid enhancer factor)/TCF (T cell factor) which serves the function of transcribing genes of the proliferative Wnt signaling pathway." (PMID: 29279096)
>
> "loss of E-cadherin releases β-catenin from the cell membrane and leads to its degradation" (PMID: 31804558)

In developmental morphogenesis, regulated downregulation of E-cadherin at adherens junctions is required for epithelial placode budding (e.g., hair follicle), mechanistically analogous to the epithelial remodeling needed in facial/eyelid/tooth morphogenesis (PMID: 25010141). Note: in BCDS, the β-catenin/Wnt connection is **mechanistically plausible and inferred** from E-cadherin biology rather than directly demonstrated in affected craniofacial tissue.

Finding 7 — Management is symptomatic and multidisciplinary

There is no disease-modifying therapy; care targets the structural manifestations:

> "Lateral tarsal strips, lower eyelid retractor disinsertion, myocutaneous advancement of the cheek and eyelids, and lateral tarsorrhaphy were performed. Correction of the ectropion, eyelid retraction, euryblepharon, and marked improvement in his lagophthalmos were achieved postoperatively." (PMID: 11476703)
>
> "Multiple surgical procedures including lateral canthoplasty, tarsorrhaphy, and midface augmentation may be necessary for correction of eyelid malposition and achieving cosmetic and functional improvements." (PMID: 39442179)

Dental rehabilitation of oligodontia uses prosthodontic/orthodontic approaches including overdentures in skeletal Class III patients (PMID: 20073547), and euryblepharon with nocturnal eyelid eversion may require surgical correction (PMID: 15220581).

Finding 8 — CDH1 allelic series with incomplete penetrance

CDH1 displays an allelic series spanning BCDS, nonsyndromic orofacial clefting, and HDGC. In nonsyndromic cleft lip/palate:

> "three probably pathogenic variants (c.760G>A [p.Asp254Asn], c.1023T>G [p.Tyr341*], and c.2351G>A [p.Arg784His]) segregated according to autosomal dominant inheritance in four nonsyndromic cleft lip with or without cleft palate (NSCL/P) families (Lod score: 5.8 at θ = 0; 47% penetrance)" (PMID: 26123647)
>
> "CDH1 contributes to NSCL/P with mainly rare, moderately penetrant variants, and CDH1 haploinsufficiency is the likely etiological mechanism" (PMID: 26123647)

Inheritance is autosomal dominant with variable expression, and many BCDS cases are de novo. The recurrent c.760G>A/p.Asp254Asn allele appears across both nonsyndromic cleft families and the BCDS spectrum and is the one shown not to predispose to cancer.

Finding 9 — Epidemiology, onset, and course

BCDS is described consistently as a "rare autosomal dominant condition" (PMID: 28301459), with fewer than ~100 reported cases, the largest molecular cohorts comprising 17 and 11 index patients. Orphanet (ORPHA:1997) assigns a prevalence of <1/1,000,000. Malformations are congenital/present at birth and detectable prenatally:

> "This case report describes the prenatal diagnosis of the extremely rare Blepharo-Cheilo-Dontic syndrome." (PMID: 41563506)

Features are structural and static rather than progressive, and life expectancy is normal (absent HDGC-class cancer risk).

Finding 10 — Comparative biology and conservation

CDH1 (human; NCBI Gene 999; HGNC:1748) has deeply conserved orthologs: mouse Cdh1 (NCBI Gene 12550), zebrafish cdh1, and Xenopus. BCDS modeling has used zebrafish (dominant-negative CDH1), mouse conditional Cdh1/Ctnnd1 deletions, Ctnnd1 knockout mice (CL/P), and Xenopus CRISPR/Cas9 CTNND1 targeting. In veterinary genetics, CTNND1 was investigated as a positional candidate for congenital entropion in Swiss White Alpine sheep, but the identified variant was deemed unlikely causal:

> "Pathogenic variants in human CTNND1 cause blepharocheilodontic syndrome 2, a rare disorder including eyelid anomalies" (PMID: 31945208)

Finding 11 — CDH1 cancer-risk management (IGCLC) for HDGC-class variants

For germline CDH1 pathogenic variants in the HDGC context, cumulative diffuse gastric cancer incidence is ~40% by age 80 and lobular breast cancer ~55% in women (PMID: 36700973):

> "Those affected show an incidence of gastric cancer of around 40% up to the age of 80 years and affected women show an incidence of 55% for the occurrence of lobular breast cancer." (PMID: 36700973)

The 2020 IGCLC guidelines provide management options directly applicable to BCDS families carrying HDGC-class CDH1 alleles:

> "Prophylactic total gastrectomy remains the recommended option for gastric cancer risk management in pathogenic CDH1 variant carriers. However, there is increasing confidence from the IGCLC that endoscopic surveillance in expert centres can be safely offered to patients who wish to postpone surgery" (PMID: 32758476)

Report by Template Section

1. Disease Information

BCDS1 is an ultra-rare autosomal dominant congenital malformation syndrome affecting eyelid, lip/palate, and dental development, caused by CDH1 variants. **Identifiers:** MONDO:0054740; OMIM #119580; Orphanet ORPHA:1997; related type BCDS2 (CTNND1). **Synonyms:** BCDS1, blepharo-cheilo-dontic syndrome 1, CDH1-related blepharocheilodontic syndrome, Elschnig syndrome. Information is derived from **aggregated disease-level resources** (OMIM, Orphanet) and from **individual patient case reports/series** (the condition is too rare for registry-scale EHR data).

2. Etiology

Primary cause: genetic — heterozygous pathogenic CDH1 variants (BCDS1); CTNND1 truncating variants (BCDS2). **Genetic risk factors:** the causal variant itself (dominant); modifier background explains variable expression. **Environmental risk/protective factors:**

none established — this is a Mendelian structural malformation syndrome. **Gene–environment interactions:** none documented. No infectious, toxic, or lifestyle contributors are known.

3. Phenotypes

Core phenotypes (congenital; physical manifestations/clinical signs; mostly stable/non-progressive):

Phenotype	Type	Suggested HPO	Frequency
Lower eyelid ectropion	Clinical sign	HP:0000656	Common/defining
Euryblepharon	Physical malformation	HP:0100660	Common
Lagophthalmos	Clinical sign	HP:0100698	Common
Distichiasis	Physical malformation	HP:0002225	Frequent
Cleft lip / cleft palate	Physical malformation	HP:0410030 / HP:0000175	Defining
Oligodontia	Dental abnormality	HP:0000677	Near-universal (mean 13.7 missing teeth)
Conical/tapered teeth	Dental abnormality	HP:0000698	Frequent
Taurodontism	Dental abnormality	HP:0000679	Reported
Dental malocclusion (Class III)	Physical	HP:0000689	P=0.001 vs controls
Ectodermal dysplasia features	Physical	HP:0000968	Reported

Quality-of-life impact: functional (ocular surface exposure/keratopathy risk from lagophthalmos; feeding/speech from cleft; mastication/esthetics from oligodontia) and psychosocial (facial dysmorphism). CTNND1/BCDS2 may add cardiac, limb, neurodevelopmental, and genitourinary burden.

4. Genetic/Molecular Information

Causal gene: CDH1 (16q22.1; HGNC:1748; NCBI Gene 999; OMIM *192090). Variant types: predominantly missense in the extracellular cadherin repeats (e.g., p.Asp254Asn/c.760G>A; p.Asp400Asn/c.1198G>A), acting dominant-negatively; CTNND1 variants are truncating (LoF/haploinsufficiency). Classification: pathogenic/likely pathogenic per ACMG/AMP. Allele frequency: BCDS alleles are rare/absent in gnomAD. Origin: germline; frequently de novo. Functional consequence: dominant-negative destabilization of the cadherin–catenin adhesion complex and accelerated E-cadherin degradation (for some missense alleles). Modifier genes: CTNND1 and the shared complex explain genetic heterogeneity; specific modifiers of expressivity are undefined. Epigenetics / chromosomal abnormalities:** not a feature of CDH1-BCDS (incidental 17q12 microdeletion in one complex pedigree is unrelated to BCDS causation).

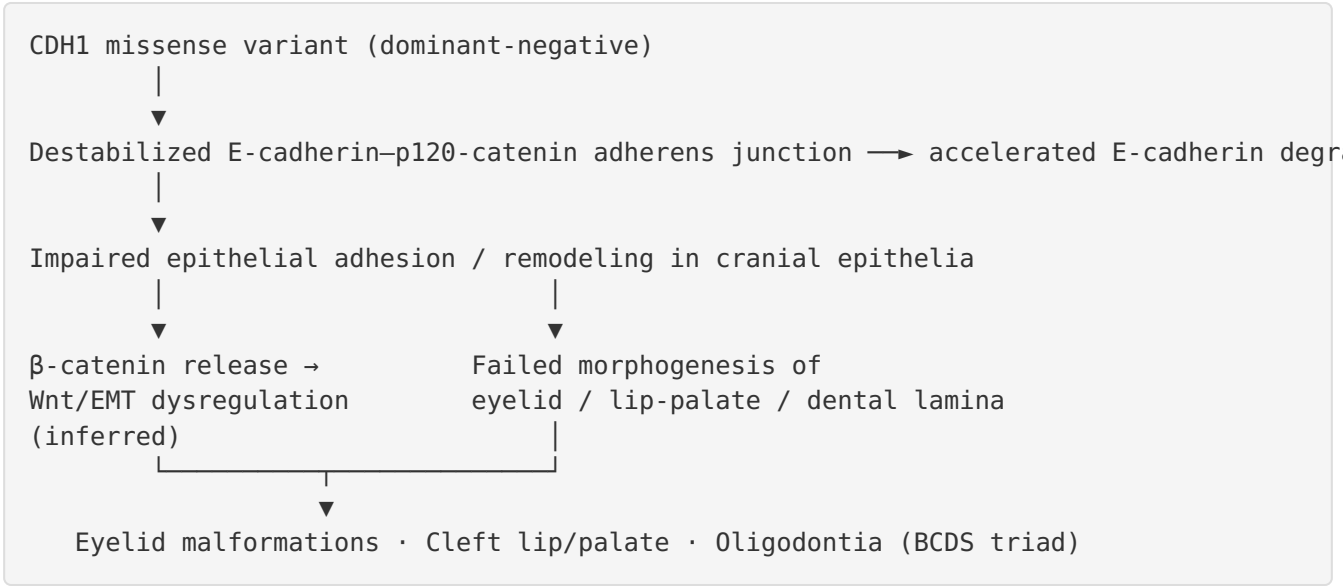
5. Environmental Information

No environmental, lifestyle, or infectious contributors are established or expected for this Mendelian disorder.

6. Mechanism / Pathophysiology

Causal chain (initiating lesion → clinical manifestation):

1. A heterozygous **CDH1 missense variant** (germline) produces a mutant E-cadherin that **leads to** a dominant-negative protein incorporated into adherens junctions (and, for some alleles, accelerated E-cadherin degradation). *(Demonstrated in zebrafish/human cells.)*
2. This **results in** destabilization of the **E-cadherin–p120-catenin (CTNND1) adhesion complex** and impaired epithelial cell–cell adhesion. *(Demonstrated.)*
3. Impaired adhesion in cranial epithelia **leads to** defective epithelial integrity, turnover, and morphogenetic remodeling (budding/fusion) in the developing **eyelid, lip/palatal shelves, and dental lamina**. *(Supported by conditional Cdh1/Ctnnd1 deletion and Ctnnd1-KO animal models.)*
4. In parallel/branch, loss of membrane E-cadherin **releases β-catenin**, perturbing Wnt signaling and EMT balance required for epithelial placode budding and fusion. *(Inferred from E-cadherin biology; not directly shown in affected human craniofacial tissue.)*
5. These developmental failures **result in** the clinical triad: eyelid malformations (failed lid morphogenesis), cleft lip/palate (failed shelf fusion), and oligodontia/tooth-shape anomalies (disrupted odontogenesis). *(Observed clinically.)*



Pathways/processes (GO): cell–cell adhesion (GO:0098609), adherens junction organization (GO:0034332), canonical Wnt signaling (GO:0060070), EMT (GO:0001837), odontogenesis (GO:0042476), palate development (GO:0060021), eyelid development in camera-type eye (GO:0061029). **Cell types (CL):** epithelial cell (CL:0000066), odontogenic/oral epithelial cell, periderm. **Subcellular (GO CC):** adherens junction (GO:0005912), plasma membrane (GO:0005886), catenin complex (GO:0016342).

7. Anatomical Structures Affected

Organ level: eyelids (UBERON:0000158), lip (UBERON:0001833), palate (UBERON:0001716), teeth (UBERON:0001091). **Body systems:** integumentary/ectodermal, craniofacial skeleton, visual adnexa. BCDS2 (CTNND1) secondarily involves heart (UBERON:0000948), limbs, kidney/urinary tract (duplex kidney; ureterocele), uterus, and CNS. **Tissue/cell:** epithelial tissue (adherens-junction-dependent), oro-palatal and dental-lamina epithelium. **Laterality:** clefts and eyelid anomalies may be unilateral or bilateral/asymmetric.

8. Temporal Development

Onset: congenital; prenatally detectable. **Pattern:** static structural malformation — non-progressive. **Critical period:** embryonic craniofacial morphogenesis (facial-shelf fusion, eyelid formation, odontogenesis). **Duration:** lifelong structural features; no spontaneous remission; outcomes depend on surgical/dental correction. HDGC-class cancer risk, when present, has adult-onset age-dependent penetrance.

9. Inheritance and Population

Inheritance: autosomal dominant, variable expressivity, frequently de novo. **Penetrance:** incomplete (~47% for CDH1-related clefting in NSCL/P families; nonpenetrant carriers reported). **Prevalence:** <1/1,000,000 (Orphanet); <~100 published cases. **Sex ratio:** no strong skew reported. **Founder/consanguinity:** not prominent; recurrent c.760G>A allele noted across families. **Carrier frequency:** not applicable (dominant).

10. Diagnostics

Clinical: recognition of the triad on physical/ophthalmologic/dental exam; dental panoramic radiography quantifies oligodontia and taurodontism. **Genetic testing (confirmatory):** single-gene CDH1 testing, multigene cleft/ectodermal panels, or WES/WGS; CTNND1 testing for BCDS2; Sanger confirmation and segregation analysis. Prenatal diagnosis is feasible via molecular testing and reverse phenotyping on fetal ultrasound. **Differential diagnosis:** isolated lagophthalmos syndrome (continuum), other oligodontia syndromes (hypohidrotic ectodermal dysplasia, Axenfeld-Rieger, Witkop, Ellis-van Creveld, oculofaciocardiodental), nonsyndromic CL/P. **Screening:** cascade genetic testing of at-risk relatives; for HDGC-class CDH1 alleles, consider upper-GI endoscopic surveillance and breast MRI (see §12).

11. Outcome/Prognosis

Survival/life expectancy: normal in the absence of HDGC-class cancer risk. **Morbidity:** functional (ocular exposure/keratopathy; feeding/speech from cleft; mastication/esthetics from oligodontia) and psychosocial. **Complications:** corneal exposure from lagophthalmos; cleft-related feeding/speech/ENT sequelae; in BCDS2, cardiac/neurodevelopmental morbidity. **Prognostic factors:** specific CDH1 variant (cancer risk), severity/extent of malformation, access to multidisciplinary surgical/dental care.

12. Treatment

No disease-modifying therapy exists; care is **symptomatic and multidisciplinary** (NCIT concepts: surgical procedure; oculoplastic/eyelid reconstruction; cleft lip/palate repair; dental prosthesis/overdenture; orthodontic therapy; supportive care). - **Eyelid:** lateral tarsal strips, lower-lid retractor disinsertion, myocutaneous cheek/eyelid advancement, lateral tarsorrhaphy, lateral canthoplasty, midface augmentation; lubrication for lagophthalmos. - **Cleft:** staged surgical repair; speech therapy; ENT/audiology. - **Dental:** prosthodontic rehabilitation (overdentures, implants when appropriate); orthodontics for Class III malocclusion. - **Cancer-risk management (variant-specific, HDGC-class alleles only):** per

IGCLC — prophylactic total gastrectomy or expert-center endoscopic surveillance; annual breast MRI from age 30 in women. No pharmacotherapy, gene therapy, RNA therapy, or immunotherapy is indicated for the malformation syndrome itself.

13. Prevention

Primary prevention: not possible (Mendelian). **Genetic counseling** is central: autosomal dominant 50% transmission risk, de novo possibility, incomplete penetrance/variable expressivity, and variant-specific cancer counseling. **Reproductive options:** prenatal diagnosis and preimplantation genetic testing for known familial variants. **Secondary/tertiary prevention:** early surgical/dental intervention to limit functional complications; for HDGC-class alleles, cancer surveillance/prophylaxis. **Cascade screening** of relatives.

14. Other Species / Natural Disease

Human CDH1 (NCBI Gene 999) is orthologous to mouse Cdh1 (NCBI Gene 12550), zebrafish cdh1, and Xenopus cadherin. CTNND1 was explored as a candidate for **congenital entropion in Swiss White Alpine sheep**, though the specific variant was judged unlikely causal (PMID: 31945208). The cadherin–catenin complex is deeply evolutionarily conserved, underpinning cross-species modeling. No established naturally occurring BCDS-equivalent companion-animal disease is catalogued; no zoonotic relevance.

15. Model Organisms

- **Zebrafish:** dominant-negative CDH1 expression reproduces adhesion defects (PMID: 29348693).
- **Mouse:** conditional Cdh1 and Ctnnd1 deletions induce BCD-reminiscent features (PMID: 28301459); Ctnnd1 knockout recapitulates the CL/P spectrum (PMID: 29805042).
- **Xenopus:** CRISPR/Cas9 CTNND1 targeting links p120-catenin to epithelial/mesenchymal craniofacial roles (PMID: 32196547).
- **In vitro:** human cell adhesion assays demonstrate dominant-negative effects.

Phenotype recapitulation: cleft lip/palate and craniofacial features are well-reproduced in Ctnnd1-KO mice; adhesion-complex dysfunction is reproduced in zebrafish/cells. **Limitations:** the full human triad (particularly the eyelid and oligodontia spectrum) is not completely recapitulated in any single model; most models use conditional/complete LoF rather than the human dominant-negative missense mechanism.

Mechanistic Model / Interpretation

BCDS1 is best understood as a **disorder of the epithelial adherens junction** during craniofacial morphogenesis. The unifying lesion is destabilization of the **E-cadherin–p120-catenin complex**, reachable by two genetic routes: dominant-negative CDH1 missense alleles (BCDS1) or truncating CTNND1 alleles (BCDS2). The convergence of two genes encoding direct binding partners onto one clinical phenotype is the strongest evidence that adhesion-complex integrity — not a single gene's dose — is the critical variable. The downstream β-catenin/Wnt/EMT arm is biologically coherent and supported by cancer and developmental literature, but in BCDS it remains an **inference** rather than a tissue-level demonstration.

Feature	CDH1 (BCDS1)	CTNND1 (BCDS2)	HDGC (CDH1)
Typical variant type	Missense (extracellular repeats)	Truncating	LoF (truncating/splice/deletion)
Mechanism	Dominant-negative	Haploinsufficiency/LoF	Loss of function (+ 2nd hit)
Core phenotype	Eyelid/cleft/dental triad	Triad + cardiac/limb/NDD/GU	Diffuse gastric + lobular breast cancer
Cancer risk	Variant-dependent, contested	Not established	High (GC ~40%, LBC ~55%)
Penetrance	Incomplete (~47% clefting)	Variable	Incomplete, age-dependent

Evidence Base

PMID	Role in report	Support / Challenge
29348693	Dominant-negative CDH1/CTNND1; triad definition	Core support
28301459	CDH1 missense + CTNND1 truncating; animal models	Core support
28181393	Quantified dental/craniofacial phenotype	Support
32302040	First DGC in BCDS	Support (cancer risk)
39596675	c.760G>A cleft without cancer	Challenge (variant-dependent)
32483297	Confirmation of cancer risk	Support
29805042	Ctnnd1-KO CL/P model	Support (mechanism)
32196547	Expanded CTNND1 phenotype; Xenopus	Support (BCDS2)
40028877	Novel genitourinary findings	Support (BCDS2 expansion)
29279096	E-cadherin/β-catenin/Wnt biology	Support (mechanism)
31804558	E-cadherin loss → β-catenin/EMT	Support (mechanism)
26123647	AD inheritance, 47% penetrance, haploinsufficiency	Support (genetics)
11476703	Eyelid reconstructive surgery	Support (treatment)
39442179	Periocular surgical management	Support (treatment)
20073547	Dental/overdenture management	Support (treatment)
41563506	Prenatal diagnosis, rarity	Support (epidemiology)
31945208	Comparative (sheep CTNND1)	Support (comparative)
32758476	IGCLC HDGC guidelines	Support (management)
36700973	HDGC lifetime cancer risks	Support (risk quantification)
36552944	Novel CDH1 p.Asp400Asn	Support (variant spectrum)
15930907	BCDS/lagophthalmos continuum	Support (differential)
38132417	Oligodontia syndrome differential	Support (differential)

Limitations and Knowledge Gaps

1. **Small evidence base.** With <~100 reported cases and largest cohorts of 11–17 index patients, phenotype frequencies and genotype–phenotype correlations are imprecise; "frequencies" are qualitative.
2. **Cancer risk unresolved.** Whether, and which, CDH1-BCDS variants confer HDGC-class risk is genuinely contested. The field lacks a validated rule linking a BCDS variant to cancer penetrance; management defaults to variant-specific counseling.
3. **β-catenin/Wnt/EMT arm is inferred.** The downstream Wnt/EMT dysregulation in affected human craniofacial tissue has not been directly demonstrated; it is extrapolated from cancer and developmental models.
4. **Mechanistic mismatch in models.** Most animal models use conditional/complete loss of function rather than the human dominant-negative missense mechanism, limiting fidelity. No single model recapitulates the complete human triad.
5. **Penetrance/expressivity modifiers unknown.** Incomplete (~47%) penetrance and nonpenetrant carriers imply uncharacterized modifiers.
6. **Quality-of-life and outcome data are anecdotal**, drawn from case reports rather than systematic patient-reported-outcome instruments.

Proposed Follow-up Experiments / Actions

1. **Variant-resolved cancer-risk registry.** Pool international BCDS cases by exact CDH1 variant and prospectively track gastric/breast cancer to define per-allele penetrance and settle the surveillance question.
2. **Functional triage assay.** Develop a standardized cell-based assay (adhesion, E-cadherin degradation kinetics, β-catenin localization) to classify each BCDS CDH1 variant as "HDGC-like LoF" vs "dominant-negative, cancer-neutral," informing counseling.
3. **Faithful disease model.** Generate knock-in mice/zebrafish carrying patient-specific dominant-negative CDH1 missense alleles (not conditional KO) to test whether they reproduce the full eyelid–cleft–dental triad.

4. **Direct test of the Wnt/EMT branch.** Use patient-derived iPSC craniofacial/dental organoids to measure β-catenin/Wnt and EMT activity, confirming or refuting step 4 of the causal chain in human tissue.
5. **Genotype–phenotype and modifier study.** WGS of BCDS families with discordant expressivity to identify modifier loci explaining incomplete penetrance.
6. **Standardized multidisciplinary outcome measures.** Apply validated ophthalmic, speech, dental, and PRO instruments across a multi-center cohort to quantify functional/QoL burden and benchmark surgical outcomes.

Evidence source legend: human clinical (case series/reports, cohorts), model organism (zebrafish, mouse, Xenopus), in vitro (human cell adhesion assays), computational (ACMG/bioinformatic variant classification). Ontology suggestions (HPO/GO/CL/UBERON/NCIT/MONDO) are provided inline for knowledge-base ingestion.
